Identification and Observation of Spatiotemporal Evolution Features of Coalfield Fire Areas Using Landsat Time-Series Decomposition

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***Abstract*—Globally uncontrolled and persistent coal fires cause substantial resource depletion and long-term environmental and safety risks. However, their long-term detection is hindered by complex subsurface dynamics, combustion behavior, and insufficient data from traditional monitoring methods. This study developed an integrated framework for the dynamic characterization of fire hazards. The approach combines time-series decomposition, abrupt change analysis, and robust regression, using Landsat-derived land surface temperature (LST) time-series data (2013–2024) for the Miquan coalfield, Xinjiang, China. The LST was first retrieved from Landsat thermal infrared (TIR) imagery via the radiative transfer equation (RTE) model. A hybrid seasonal-trend decomposition using loess (STL) and breaks for additive season and trend (BFAST) model then separated trend, seasonal, and residual components, enabling the detection and quantification of abrupt LST changes related to coal-fire dynamics. Thermal trends and anomalies were further quantified using the Mann–Kendall (MK) test, Sen's slope, and a novel average annual deviation (AAD) metric. Crucially, using random sample consensus (RANSAC)-based regression, persistent potential fire pixels were extracted and categorized into four frequency regimes, with high robustness to noise and data gaps. The resulting coal-fire frequency maps achieved over 77% spatial accuracy relative to field records, demonstrating enhanced continuity and reliability in long-term coal-fire monitoring. The framework provides a versatile, pixel-level, and analytical paradigm for understanding the spatiotemporal evolution of these challenging landscape phenomena.**

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I. Introduction

COALFIELD fire areas refer to regions affected by the combustion of underground or surface coal seams, which develop when favorable ignition conditions are met and often display clear spatial and temporal scales [1]. These fires represent a unique category of persistent combustion phenomena. Distinct from wildfires, which are typically characterized by marked seasonality and regionality, coalfield fires can endure for weeks, months, or even decades [2]. Their initiation and spread are primarily governed by coal oxidation and spontaneous combustion processes, and such events are particularly prevalent under conducive natural conditions or in areas disturbed by intensive mining activities [3].

Due to their concealed nature, long duration, and uncontrollable behavior, coal fires are widely recognized as a global environmental and geological disaster. They are widespread in coal-rich countries with extensive mining operations, including China, USA, India, Australia, and South Africa [4]. Prolonged large-scale coal seam combustion results in the irreversible loss of valuable coal resources and triggers a severe cascade of environmental and geological impacts. These consequences include severe atmospheric, hydrological, and pedological contamination [5], [6], extensive land subsidence and geological collapse [7], and the emission of substantial greenhouse gases, which intensifies global climate change [8]. Collectively, these threats pose substantial risks to human health, industrial safety, ecological integrity, and socioeconomic stability across the affected regions [9].

Traditional approaches to coal-fire detection have primarily relied on surface and subsurface measurement techniques. These techniques include the identification and localization of active fires through geo-environmental indicators such as gas composition, temperature anomalies, geomagnetic variations, electric potential differences, and the mapping of subsurface cavities within and around active fire areas [2], [10]. While such ground-based geophysical and geochemical technologies have demonstrated notable effectiveness through decades of practical application and detection [11], their utility remains severely constrained. The limitations chiefly stem

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from topographic complexity, temporal limitations, and the performance characteristics of survey instruments [12]. Moreover, successful deployment typically necessitates extensive field campaigns, which are not only time-intensive and labor-demanding but also expose investigators to considerable safety risks in hazardous coal-fire environments [13]. By contrast, remote sensing enables the systematic characterization for the relatively safe and efficient detection and monitoring of coal-fire dynamics over large and often inaccessible regions. The clear advantages of this approach include broad spatial coverage, straightforward data acquisition, and high temporal resolution [14].

Remote sensing imagery provides a robust means of capturing diverse anomalous responses induced by (sub)surface combustion. By extracting and integrating single or multiple indicators and applying suitable interpretation techniques, it is possible to effectively discriminate fire-affected areas from the surrounding background. Current remote sensing observations of coal fires are generally categorized into three principal types: surface thermal anomalies [15], surface deformation [16], and changes in the overburden and vegetation cover [17]. Among these methods, the detection of surface thermal anomalies constitutes a crucial step in the accurate identification and delineation of coal-fire areas. Such anomalies are primarily monitored using thermal infrared (TIR) and shortwave infrared (SWIR) sensors [14]. Methods for extracting coal-fire-related thermal anomalies generally fall into two broad categories: threshold-based approaches (which can be manual [18], [19], semi-automatic [20], [21], or fully automated [22], [23]) and nonthreshold-based approaches (including spatial statistics [24], [25], image processing [26], [27], and pattern recognition [28], [29]). Substantial progress has been achieved through the integration of quantitative remote sensing with geographic information systems (GISs), particularly in revealing the spatiotemporal dynamics of coal fires and supporting fire risk and environment assessment [30]. Furthermore, long-term remote sensing observations have demonstrated distinct advantages in objectively quantifying coal-fire risks [31], including their location [32], scale and intensity [33], [34], and safety evaluation [35]. This longitudinal monitoring also provides valuable insights into fire behavior [36], enabling the prediction of changes in fire regimes such as expansion or reduction [37], shifts in distribution and spread [38], and broader migration trends [39].

In land surface temperature (LST) research, the construction of spatiotemporally continuous observation fields has enabled researchers to conduct deeper analyses of surface thermodynamic characteristics. This shift facilitates precise trend recognition, pattern parsing, and the capture of subtle dynamics within large-scale datasets, establishing LST as a critical indicator for understanding energy balance amidst global change [40]. When examining surface dynamics, LST time series are typically decomposed into three primary components: long-term trends reflecting climate change; seasonality produced by the annual temperature cycle; and abrupt changes (mutations) caused by natural or anthropogenic disturbances such as wildfires, extreme weather, or rapid urbanization [41]. Previous studies frequently relied on interannual data to characterize these trends, but these often ignore intrayear variability and fail to provide the specific timings of mutations [42]. Thus, utilizing high-frequency intra-annual sequences (e.g., eight-day resolution) to simultaneously capture these components has become the dominant direction for precise thermal environment analysis [43].

The methodology for detecting mutations in LST time series follows two primary paradigms. The first involves unknown component forms, such as seasonal-trend decomposition procedure based on loess (STL) and detecting breakpoints and estimating segments in trend (DBEST) method [44]. While these offer flexibility for nonlinear trends, they remain sensitive to noise and struggle with missing satellite data. The second paradigm utilizes known component forms, represented by algorithms like breaks for additive season and trend (BFAST) and the Bayesian ensemble algorithm for change point detection, Seasonalities, and Trends (BEAST) [45]. Notably, the BEAST algorithm employs a Bayesian model averaging (BMA) scheme to synthesize numerous competing models, demonstrating superior robustness in handling missing values and reducing model misspecification risks [41]. For identifying significant trajectories within these complex datasets, robust statistical measures like the Theil–Sen estimator and Mann–Kendall test are standard for mitigating outlier impacts [46].

In application, LST dynamics are extensively utilized to monitor surface urban heat islands (SUHIs) [47], heatwaves [48], and drought-related soil moisture status [49]. Simultaneously, the integration of deep learning (DL) is steering thermal anomaly monitoring toward an intelligent era, facilitating a shift from traditional static mapping toward near-real-time monitoring [50]. A pivotal advancement in this transition is the move from rigid, empirical thresholds to adaptive, "environment-aware" monitoring. For instance, Kang et al. [51] integrated vegetation and atmospheric variables into DL architectures to improve generalization in heterogeneous landscapes. This evolution toward quantitative assessment is further exemplified by Zhang et al. [52], who combined the BFAST algorithm with a Random Forest model to capture vegetation mutations caused by underground spontaneous combustion in coal waste dumps, achieving early quantification of heat sources via the spontaneous combustion intensity index.

These multidimensional approaches have evolved into multitask frameworks that optimize fire detection, severity classification, and pixel-level segmentation simultaneously [53]. Structurally, the focus has shifted from standard convolutional neural networks (CNNs) toward vision transformers (ViTs) and advanced contextual models, such as FlameViT [54] and Swin-Unet [55]. These models capture global dependencies within multispectral imagery to delineate fire fronts and smoke plumes with unprecedented precision. When combined with robust statistical frameworks like BMA to manage model uncertainty and inherent data gaps, these technological developments provide a comprehensive foundation for global wildfire management and real-time thermal hotspot surveillance [40].

Despite the notable advancements in remote sensing applications, most previous studies on coal-fire monitoring have

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highlighted that insufficient, inconsistent, or incomplete image datasets exacerbate biases and misidentifications when recognizing thermal anomaly pixels influenced by random environmental disturbances [56]. Equally critical are the temporal span and observation frequency of satellite imagery, which directly affect the accuracy of coal-fire detection and extraction [57]. However, the majority of prior research has been restricted to short-term monitoring, often relying on single-scene images or relatively short time series [58]. Such limitations fundamentally impede a comprehensive understanding of the spatiotemporal evolution of coal fires, particularly their continuity, fluctuations, and abrupt changes throughout the entire fire life cycle.

In response to these challenges, recent studies have progressively adopted novel analytical frameworks for quantitative coal-fire monitoring based on extended time series. For instance, Mujawdiya et al. [59] applied the STL model to nearly two decades of MODIS LST time series at the pixel level (928 m). They successfully identified fire-affected pixels by extracting anomalies from the long-term trend component. Building on this work, and with the advent of higher resolution sensors, they later proposed a context-based detection method that utilized nearly a decade of diurnal and nocturnal VIIRS (750 m) LST observations [60]. By analyzing deviations between daytime and nighttime trends, this approach enabled the identification of coal fires across broader spatial extents. At an even finer scale (30 m), Chen et al. [58] employed pixel-level time-series analysis using 35 years of Landsat LST data to delineate coal fires in the Wuda coalfield of China with improved precision, thereby revealing their sustained dynamic evolution over multiple decades. Existing long-term time-series investigations have largely been confined to the singular application of STL decomposition, with limited exploration of alternative analytical methodologies for remote sensing time series. This reliance may have restricted the capacity to detect abrupt changes (breakpoints) and to interpret anomalies effectively, thereby limiting the isolation of true coal-fire signals from background noise.

To overcome these methodological limitations, this study develops a comprehensive time-series decomposition and trend detection framework grounded in breakpoint modeling. By systematically separating seasonal, trend, and residual components, the framework enables the precise identification of abrupt changes in coal-fire dynamics. Specifically, it integrates the concept of average annual deviation (AAD) with the random sample consensus (RANSAC) model, thereby significantly enhancing robustness against outliers and noise. Through this combined approach, the framework systematically reveals the spatiotemporal abruptness, intensity evolution, and distinction between anomalous signals and background in coal-fire activity. This study thus provides a novel methodological pathway for the long-term dynamic monitoring and risk assessment of coalfield fires.

II. Study Area and Data

A. Study Area

The Miquan Coalfield is located in the southern part of Midong, Ürümqi, within the Xinjiang Uygur Autonomous Region, China, extending between 87°42′36″–87°52′03″E and 43°53′14″–43°58′49″N. Geographically, the coalfield lies along the southern margin of the Junggar Basin and is characterized by a typical continental arid to semi-arid climate. The region forms part of the Bogda Foothills, where the dominant landforms are tectonic–erosional in origin, with surface elevation ranging from 813 to 964 m. Structurally, the area is defined by a northwesterly dipping monocline, with strata generally inclined between 79° and 80°. Within the Miquan Coalfield, the principal fire-affected zones can be subdivided into nine functional sections: coal mining activity areas (sections B, D, E, F, and H), coal storage and production areas (sections G and I), and coal-fire-suppression areas (sections A and C).

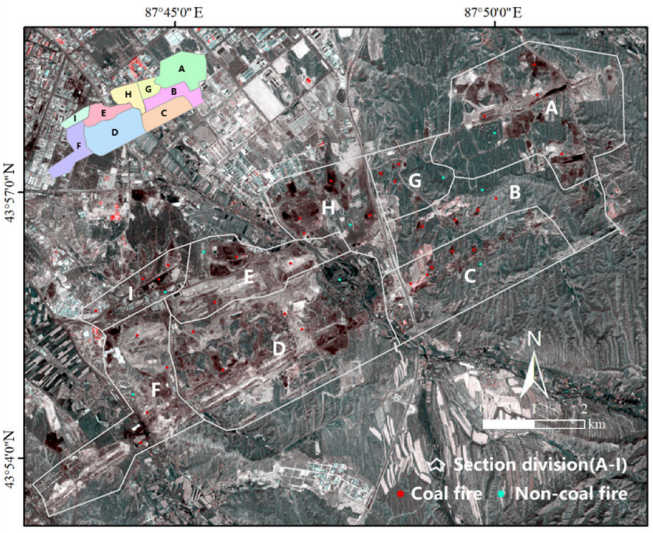


Fig. 1. Location of the study area: Miquan Coalfield, subarea division (Sections A–I), and distribution of coal-fire and nonfire sample points. The background map is a Sentinel-2 standard natural color composite (B2–B4) image from May 14, 2017, at 10-m spatial resolution.

For this study, historical coal-fire sample points were meticulously selected from different subsections of the fire-affected zones, encompassing representative fire types such as surface, underground, outcrop, and abandoned mine fires (Fig. S1). To enable effective comparison, noncoal-fire sample points with diverse surface characteristics—including vegetation, bare ground, gravel, and soil—were simultaneously identified as environmental background References. These two categories of sampling points provide a clear contrast between coal-fire combustion signatures and baseline environmental conditions in the study area. The spatial divisions of the coalfield, along with the locations and distributions of the sampling points, are illustrated in Fig. 1.

B. Remote Sensing Data

This study employed synergistic observations from the Landsat-8 satellite (launched in 2013) and the Landsat-9 satellite (launched in 2022). Both satellite platforms carry two key sensor payloads: the Operational Land Imager (OLI), which provides 30-m resolution visible to near-infrared spectral data (bands B2–B7); and the TIR Sensor (TIRS), which acquires

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100-m resolution TIR data (bands B10–B11). For consistency in the analysis, all band data were resampled to 30 m. Operating together, the twin satellites achieve an effective revisit interval of 8 days (16 days for an individual platform). Through the Google Earth Engine (GEE) platform, a total of 622 Landsat-8/9 images (Level-1 Terrain Precision) covering the Miquan Coalfield (Path/Row = 143/030) were obtained for the period January 2013 to December 2024. They have undergone rigorous geometric terrain correction and orthorectification. The temporal distribution of these acquisitions is illustrated in Fig. S2. Firstly, only images with less than 20% cloud cover were retained. Subsequently, each image was clipped to the vector boundary of the study area and reprojected to the WGS84 UTM Zone 45 N coordinate system, thereby standardizing the entire dataset for subsequent processing and time-series analysis. The preprocessing procedure is detailed in Appendix I (see the Supplementary Material) and comprises several key steps to ensure data quality and computational efficiency. A dual-stage atmospheric correction strategy—integrating FLAASH-based spectral features and localized AOD data—was implemented to ensure the physical consistency of the subsequent trend analysis.

C. Supplementary Data

To comprehensively validate the coal-fire monitoring results and support the modeling framework, this study integrated multiple auxiliary datasets. Ground-truth observations were primarily derived from previous field investigations [9], [25]. The principal parameters collected included surface temperature, active fire-point coordinates, and fire zone boundaries and extents. Surface temperature measurements of active fire spots were obtained using TIR cameras (FLIR T-610) and handheld pyrometers (ZyTemp-TN43), in conjunction with GPS receivers (eTrex-201x) for accurate hotspot localization and identification. Additional information on fire-point locations and combustion status for 2015 and 2016 was provided by professional surveys conducted by the Xinjiang Fire Extinguishing Engineering Bureau [61]. Data covering 2018–2020 were obtained through multiple field campaigns across representative coal-fire anomaly areas, supplemented by our earlier research findings [62]. Furthermore, historical meteorological records were collected from the Miquan District Meteorological Station for the period 2013–2020. These datasets—comprising atmospheric temperature, pressure, humidity, and elevation parameters—served as benchmark inputs for the atmospheric correction of the remote sensing imagery and for LST inversion modeling.

For remote sensing validation, the Landsat Collection 2 Level 2 (LC8-9LC2L2) LST product, derived from Landsat-8/9 Band 10 at 30-m spatial resolution, was synchronously acquired from the GEE platform. A dedicated time series of this LST data, spanning 2015–2020 (to match the field measurement period), was constructed to facilitate the accuracy assessment of the inversion algorithm. Furthermore, a spatiotemporal distribution map of thermal anomalies in the Miquan Coalfield, previously generated using the hotspot sequence frequency method applied to Landsat time-series data (2013–2020) [25], [63], was employed for cross-comparison and reliability verification of the coal-fire extent extracted in this study.

III. Methodology

This study implemented a comprehensive, fully automated image preprocessing workflow on the GEE platform. This covered Landsat OLI/TIRS data loading, topographic and geometric corrections, radiometric (atmospheric) corrections, and LST inversion (Section III-A) via the radiative transfer equation (RTE) inversion model. We generated a standardized weekly scale LST time-series product for the Miquan Coalfield (2013–2024). Detailed temporal composition and quality control are provided in Appendix II and Tables S1–S3 (see the Supplementary Material).

Subsequent analyses were primarily performed in the R programming environment. First, the temporal fluctuations in the LST series were resolved using the seasonal and trend decomposition using loess (STL) algorithm (Section III-B), which decomposed the time series into seasonal, trend, and residual components. Based on these results, the BFAST model was then applied to iteratively detect breakpoints in both the trend and seasonal components (Section III-C). This procedure quantified the magnitude and statistical significance of the trend, identified the timing of breakpoints, and characterized the type of abrupt changes in the thermal anomaly series. For the trend component of the STL decomposition after the background correction, the Mann–Kendall (MK) test in combination with Sen's slope estimator was utilized (Section III-D1 and III-D2) to assess long-term dynamics, quantify the rate of change, and evaluate statistical significance in interannual LST trends across the fire area. Additionally, an AAD metric was introduced to quantify the anomalous yearly deviation of coal-fire LST signals from background thermal conditions (Section III-D3). Finally, the random sample consensus (RANSAC) algorithm was applied (Section III-E) to fit a robust linear trend to each pixel's LST time series across the study area focusing on the background-corrected LST. Pixels exhibiting persistent anomalous values above the fit baseline trend were then identified as potential coal-fire pixels. Their spatial and temporal patterns were subsequently analyzed to evaluate coal-fire combustion dynamics, spatial distribution, spatiotemporal evolution, and key influencing factors.

Detailed methodological evaluations are provided in the Appendix (Sections III–V) (see the Supplementary Material). Each method comprises three consistent subsections: physical and statistical rationale for parameter selection, sensitivity experiment design, and quantitative evaluation. These analyses ensure that the STL, BFAST, and RANSAC configurations are grounded in physical reasoning rather than arbitrary tuning.

A. LST Retrieval

In this study, the digital numbers (DNs) acquired by Landsat sensors were first converted into at-sensor spectral radiance values through linear radiometric calibration. For the TIR bands, LST was retrieved using the RTE model [64]. This model accounts for atmospheric effects on surface thermal radiation by assuming that the total spectral radiance

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[$L_{sensor,\lambda}$, W/(m^2·sr·μm)] received by the satellite at wavelength λ comprises three components: the surface blackbody radiance $B_\lambda(T_s)$, the upwelling atmospheric radiance ($L^{\uparrow}_{atm,\lambda}$), and the downwelling atmospheric radiance ($L^{\downarrow}_{atm,\lambda}$)

$$L_{sensor,\lambda} = \left[\varepsilon_\lambda B_\lambda(T_S) + (1-\varepsilon_\lambda) L^{\downarrow}_{atm,\lambda}\right]\tau_\lambda + L^{\uparrow}_{atm,\lambda} \quad (1)$$

where ε is the land surface emissivity and τ is the average atmospheric transmittance at wavelength λ. The atmospheric parameters of $\tau, L^{\uparrow}_{atm,\lambda}$, and $L^{\downarrow}_{atm,\lambda}$ were estimated using the MODTRAN standard model in conjunction with NCEP reanalysis atmospheric profiles [65]. These calculations required initial field-measured parameters, including elevation, air pressure, temperature, and relative humidity (Section II-C).

In the process of estimating surface emissivity (ε), land cover types were distinguished using the normalized difference vegetation index (NDVI) threshold method [66]. The corresponding emissivity values were assigned based on the fractional vegetation cover (f_v) per pixel

$$\varepsilon = \varepsilon_v f_v + \varepsilon_s (1 - f_v) \quad (2)$$

where $\varepsilon_v = 0.986$ and $\varepsilon_s = 0.967$ denote the emissivity of vegetation and bare soil, respectively. The red (ρ_{red}) and near-infrared (ρ_{near}) bands used to compute NDVI were derived from atmospherically corrected OLI data. Once the emissivity is determined, the surface blackbody radiance $B_\lambda(T_s)$ can be calculated. The surface brightness temperature is then retrieved by applying the inverse form of Planck's function. The specific derivation of the LST inversion algorithm, parameterization, and application to the coal-fire area are provided in [25].

B. Time-Series Decomposition

The STL is a robust time-series decomposition method that employs locally weighted regression smoothing (Loess) [67]. STL decomposes a time series (Y_t) into three additive components: a long-term trend (T_t), a periodic seasonal (S_t), and an irregular residual (e_t)

$$Y_t = T_t + S_t + e_t \quad (t = 1, 2, \ldots, n). \quad (3)$$

The STL algorithm is implemented through iterative smoothing, which consists of two embedded loops. The inner loop updates and fits the seasonal and trend components, while the outer loop adjusts robustness weights to minimize the influence of outliers through robust iteration.

In this study, the STL function implemented in the R environment [68] was applied to decompose all LST time-series variables. The key decomposition parameters were carefully optimized for the coal-fire LST dataset.

1) Seasonal Window: $n_s = 52$ (to precisely match the weekly sampling frequency of the annual cycle).

2) Trend Window: $n_t = 80$ (in accordance with the empirical criterion of 1.5 times the cycle length).

3) Low-Pass Filter Window Length: $n_l = 53$ (set to the smallest odd period greater than the cycle length).

Crucially, robust decomposition was enabled (robust = TRUE), ensuring that thermal outliers were assigned lower weights, thereby reducing their undue influence on the overall long-term trend and seasonal patterns.

C. Breakpoint Detection

1) BFAST Model and Segmented Fitting: The BFAST method is an iterative fitting algorithm based on the additive decomposition model [69]. After decomposing the original time series, BFAST efficiently detects structural breakpoints (abrupt changes) within either the trend or seasonal components over the observation period (t). For the trend component (T_t), the time series is fit in piecewise linear segments. For each interval $t_i < t \le t_{i+1}$, with $t_0 = 0$, the linear model is expressed as

$$T_t = \alpha_i + \beta_i t \quad (i = 1, 2, \ldots, m) \quad (4)$$

where m is the total number of breakpoints in the trend component and α_i and β_i are the intercept and slope of the linear model for the ith segment, respectively.

For the seasonal component (S_t), a segmented linear seasonal model is fit for each interval $t_j < t \le t_{j+1}$, with $t_0 = 0$. The model is given by

$$S_t = \sum_{j=1}^{k} \gamma_j sin\left(\frac{2\pi jt}{f} + \delta_j\right) \quad (5)$$

where j denotes the location of seasonal breakpoints, $j = 1, 2, \ldots, q$; q is the total number of seasonal breakpoints; k is the number of harmonic terms (harmonic order); and f represents the frequency; γ_j and δ_j are the amplitude and phase coefficients (number of segments), respectively.

2) Time-Series Change Detection and Parameterization: Within the BFAST framework, the iterative detection process begins by identifying trend breakpoints on the residual term (e_t), after which the residuals are updated with a segmented linear fit to the trend (T_t). Seasonal breakpoints are then iteratively detected on the updated residuals until convergence is reached, i.e., no additional breakpoints remain in the residuals [70]. To identify structural breaks in the trend component, this study employed the moving sum of residuals (MOSUM) test [71], which is based on least-squares residuals. The MOSUM statistic is expressed as follows:

$$\text{MO}(t) = \frac{1}{\hat{\sigma}\sqrt{n}} \sum_{i=nt}^{nt+h} \hat{e}_i \quad (6)$$

where h is the moving window bandwidth (the minimum segment length in MOSUM), $\hat{\sigma}$ is the standard deviation (STD) of the residual term (e_t), and n is the number of observations in the historical phase (the length of the time series). A breakpoint is considered significant if |MO(t)| deviates from 0 beyond the 95% confidence boundary. In this study, the LST time series consists of approximately 600 observations, with 52 weeks per year, and thus, the frequency was set to $f = 52$. The minimum segmentation was defined as $h = 0.1 \times 600 = 60$ weeks, corresponding to the typical minimum duration of a coal-fire event. Seasonal terms were smoothed and fit using a harmonic model with the Fourier series harmonic order set to 3. For strongly nonsinusoidal periodic patterns, a dummy-variable fluctuation model was adopted instead. The minimum number of iterations for detecting convergence was set to 5, and the statistical significance threshold was fixed at 0.05.

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D. Trend Analysis and Anomaly Metrics

1) Mann–Kendall Trend Test: The MK test, also referred to as a distribution-free rank test, is a widely used nonparametric method for detecting the presence of monotonic trends (either upward or downward) in time-series data [72]. The MK test was implemented in R using the Kendall package. Trend existence is assessed by comparing the magnitude (sign of the difference) of each pair of data in the series using the rank correlation coefficient. For the time series with n independent observations $(x_1, x_2, \ldots, x_n)$, the test statistic S is defined as

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \operatorname{sgn}\left(x_j - x_i\right) \tag{7}$$

where the sign function $\operatorname{sgn}(\theta)$, with $\theta = x_j - x_i$, is

$$\operatorname{sgn}(\theta) = \begin{cases} 1, & (\text{if } \theta > 0) \\ 0, & (\text{if } \theta = 0) \\ -1, & (\text{if } \theta < 0). \end{cases} \tag{8}$$

When $n > 8$, the variance of S is computed as

$$\operatorname{Var}(S) = \left(n(n-1)(2n+5) - \sum_{k=1}^{q} t_k(k)(k-1)(2k+5)\right) \Big/ 18 \tag{9}$$

where t_k denotes the tied value of the kth number of the time series and q is the number of data with the same value. The standardized test statistic Z is then calculated as

$$Z = \begin{cases} (S-1) \Big/ \sqrt{\operatorname{Var}(S)} & (\text{if } S > 0) \\ 0 & (\text{if } S = 0) \\ (S+1) \Big/ \sqrt{\operatorname{Var}(S)} & (\text{if } S < 0). \end{cases} \tag{10}$$

In the two-tailed test, a positive Z indicates an upward trend, while a negative Z indicates a downward trend. The magnitude of $|Z|$ reflects the significance of the trend (significance level $\alpha = 0.05$): $|Z| < 1.96$: not significant; $1.96 < |Z| < 2.56$: significant; $|Z| > 2.56$: highly significant.

2) Sen's Slope Estimator: For each functional section (A–I), both coal-fire and nonfire (background) sample points were subjected to STL decomposition separately. The decomposition yielded seasonal, trend, and residual components for each time series, and the extracted trend components are shown in Fig. 3. To isolate the fire-induced thermal signal, the background trend was used as a regional baseline. As illustrated in Fig. 4, the net fire trend (NFT) was obtained by subtracting the background (nonfire) trend from the corresponding coal-fire trend within the same functional section at each weekly time step over the period 2013–2025. These resulting weekly NFT values were then grouped by calendar year (with up to 52 observations per year). The collection of net trend observations within each year was defined as the background-referenced annual net trend (BANT).

Sen's slope estimator was then applied to the yearly BANT series to quantify the magnitude and direction of interannual fire-related thermal evolution. The slope of the trend was estimated using the robust Sen's slope method [73], which calculates the median of all pairwise slopes in the time series.

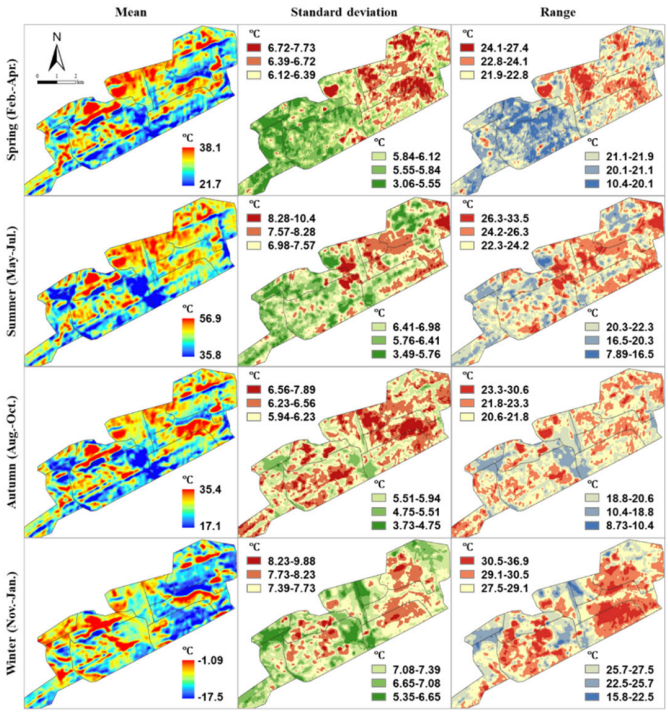


Fig. 2. Spatial distribution and seasonal statistics of LST Mean, STD, and Range across the Miquan coalfield (2013–2024). Computed independently per season to isolate fire signals from climatic fluctuations, these layers establish the baseline thermal regime. The contrast between high Mean and low STD/Range in fire areas identifies stable internal heat sources, providing the methodological basis for separating persistent combustion from transient environmental noise and surface heterogeneity.

For each pair of observations within a given year, the slope s_k is computed as

$$s_k = \left(x_p - x_q\right) \Big/ (p - q) \tag{11}$$

where x denotes the LST trend component (BANT) after background removal and x_p and x_q represent two observations in the weekly LST sequence for a given year ($1 \le q < p \le 52$, with 52 observations per year).

3) Anomalous Deviation Analysis: To further characterize the magnitude of coal-fire thermal anomalies on an annual basis, the AAD was calculated using the BANT time series.

For each year, AAD was defined as the mean absolute deviation of weekly BANT values relative to the background reference

$$\mathrm{AAD} = \left(\sum_{1}^{n} \Delta x_i\right) \Big/ n \tag{12}$$

where Δx_i denotes the weekly background-referenced net LST trend (i.e., BANT) at time step i and n is the number of valid weekly observations in that year (≤ 52). This metric enables interannual comparison of coal-fire intensity and provides a concise quantitative measure of the divergence between coal-fire thermal behavior and the regional environmental baseline.

Both the Sen's slope estimation of BANT and the computation of AAD were implemented in the R programming environment.

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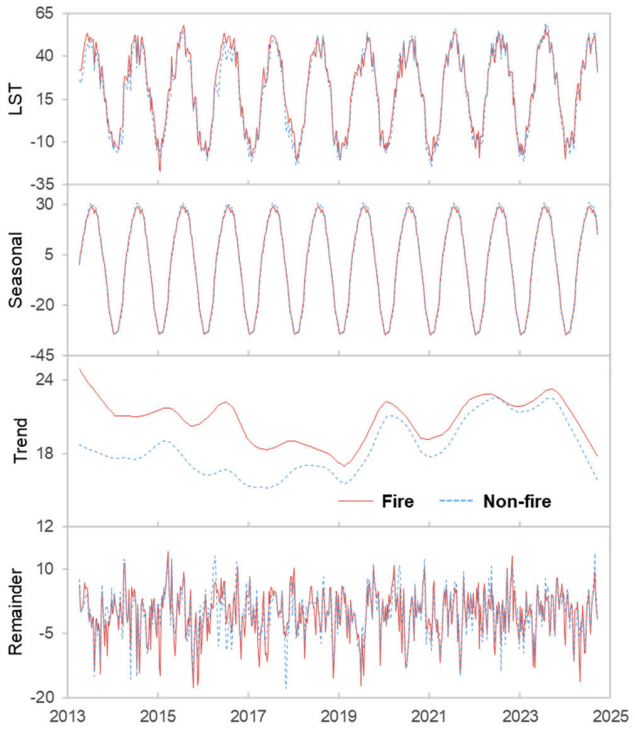


Fig. 3. STL decomposition of LST time series for coal-fire (red) and noncoal-fire (blue) sample points. Panels compare trend, seasonal, and residual components. This decomposition highlights the trend divergence essential for identifying subsurface combustion. It serves as the analytical framework for separating fire-driven thermal signals from natural climatic oscillations, establishing the basis for subsequent trend-based anomaly detection.

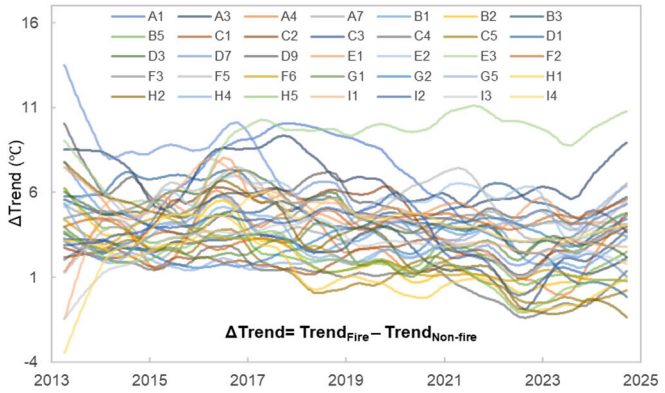


Fig. 4. Spatiotemporal distribution of the NFT, ΔTrend. By subtracting background trends from fire-point components, this metric isolates intrinsic thermal signals from shared environmental variability. It establishes a standardized framework for quantifying the net impact of combustion activity and suppression efforts, facilitating a comparative analysis of long-term fire dynamics across different management sections.

E. Fire Extraction

To robustly extract the coal-fire-induced thermal signal, the RANSAC algorithm [74] was employed. RANSAC is a widely recognized robust regression method specifically designed to fit linear models in datasets containing a high proportion of outliers, which perfectly suits the identification of subtle fire signals against varying background noise

$$y = \beta_0 + \beta_1 x + \epsilon \tag{13}$$

where β_0 and β_1 represent the model parameters (intercept and slope), x is the input data (time step), y is the background-corrected LST value, and ϵ is the residual error. The algorithm proceeds iteratively by randomly sampling minimal subsets of the data and fitting candidate models. For each data point (x_i, y_i), it is classified as an inlier if the deviation from the fit line is below a predefined threshold τ

$$|y_i - (\beta_0 + \beta_1 x_i)| \le \tau. \tag{14}$$

The model with the maximum proportion of inliers is selected as the optimal fit, and a final regression is refitted using all inliers. The maximum number of iterations N required to ensure the model achieves a desired confidence probability u is given by

$$N = \log(1-u) \big/ \log(1-p^m) \tag{15}$$

where $p = n_{inliers}/n_{total}$ is the inlier proportion and m is the minimum sample size required to determine the model parameters. This robust fitting process allows the reliable definition of the background-corrected LST baseline, enabling the subsequent identification of pixels exhibiting persistent thermal anomalies significantly deviating from this fit line.

The RANSAC fitting in this study was implemented through a custom function (ransac_fit) developed within R. The algorithm parameters were set as: a maximum iteration limit $N = 1000$, inlier residual threshold $\tau = 0.5 \sim 2$ °C, and a minimum acceptable inlier proportion of $> 50\%$. The input data were the background-referenced LST trend component derived from STL decomposition, and the outputs were the fit parameters (β_0 and β_1), representing robust estimates of the long-term LST trend characteristics for each pixel.

F. Reliability Evaluation

1) LST Validation and Consistency Analysis: The reliability of the RTE-derived LST (Section III-A) was assessed through a dual approach. First, to verify the absolute magnitude accuracy, the retrieved LST was compared with the official Landsat Collection 2 Level 2 LST product (LC8-9LC2L2) over spatiotemporally overlapping pixels. This cross-validation quantified consistency using root-mean-square error (RMSE), mean absolute error (MAE), and linear regression coefficient (R^2). Second, considering the distinct physical nature of land and air, a trend consistency analysis was conducted between the RTE-derived LST and ground-based air temperature records from Midong meteorological stations (2013–2020). While they are physically different quantities, their temporal synchronization was assessed using R^2 and Pearson's correlation coefficient (R) to verify the algorithm's seasonal stability.

2) Spatial Validation: The spatial accuracy of the final coal-fire extraction was assessed at two distinct levels. At the fire-point level, remotely detected fire points (2015–2020) were cross-checked with field records of combustion states (on fire/no fire) and their precise locations within confirmed fire areas. Accuracy was subsequently quantified by the positive

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recognition rate of fire-point location and combustion state compatibility. At the fire area level, fire extents extracted by the RANSAC method were compared with historical distribution maps (2013–2020) derived from the Hotspot Sequential Frequency Superposition (HSFS) extraction method [25], [63], and the agreement was quantified using the spatial overlap rates.

IV. Results

A. Spatiotemporal Pattern and Statistics of LST

The reliability of the thermal data foundation was rigorously evaluated through cross-validation and trend consistency analysis. As demonstrated in Table S4, the RTE-derived LST exhibits strong agreement in absolute magnitude with the official LC8-9LC2L2 products, verifying the accuracy of the retrieval algorithm. Furthermore, a trend consistency analysis was conducted by comparing the retrieved LST with ground-measured air temperatures from Midong meteorological stations (2013–2020). As illustrated in Fig. S3, the high correlation between these two variables confirms that the RTE-based inversion maintains excellent temporal consistency and seasonal stability.

Based on the weekly LST time-series data (2013–2024), the multiyear Mean, STD, and extreme temperature Range were calculated for each pixel across the study area. To mitigate the influence of seasonal fluctuations, statistics were computed independently per season following the four-season classification defined in Fig. 2, and the spatial patterns of seasonal variability were subsequently analyzed. Results reveal that the persistent and high-intensity thermal radiation from coal fires creates pronounced temperature contrasts between the combustion zones and the surrounding unaffected areas. Notably, mean temperatures in spring and autumn within the fire area are similar, with highly congruent thermal anomaly spatial patterns. On average, the active coal-fire regions are 5 °C–7 °C higher than the normal background, forming distinct hotspot clusters. Areas located within the fire zone consistently exhibit elevated mean values across all seasons, indicating a likely presence of sustained high-intensity coal combustion.

Analysis of the seasonal differentiation of temperature variability reveals a strong spatial correlation between the STD and Range of LST. These distributions are jointly regulated by seasonal dynamics and combustion stability. Across the study area, both STD and Range values reach maximum levels in winter (STD: 5 °C–10 °C; Range: 16 °C–37 °C) and decline to minimum levels in spring and autumn. Spatially, the eastern fire area exhibits generally higher STD and Range values than the western area, and the spatial pattern response of Range is broadly consistent with that of STD. Within the coal-fire affected areas, STD values typically remain below 7 °C, with the Range typically falling between 21 °C and 31 °C, whereas in nonfire (background) areas, range levels are slightly lower (20 °C–28 °C). High STD or Range values observed outside the fire areas are mainly attributable to land cover heterogeneity and seasonal climate fluctuations. Crucially, the persistent combustion in active fire areas leads to a relatively stable and homogeneous spatial distribution of LST. This distinction demonstrates that the persistent internal heat source results in natural LST oscillations with smaller spatiotemporal variability in fire-affected areas. These areas are characterized by high mean temperatures but lower interannual fluctuations.

B. STL Trend Deviation of Coal Fire

The initial step in analyzing the thermal dynamics involved applying the STL algorithm to both the coal-fire sample points and the noncoal-fire background points, to extract and compare their respective trend components. The decomposition applied to both sample point sets demonstrates that the trend, seasonal, and residual components each exhibit deviations of varying magnitude (Fig. 3). Because the background points were rigorously screened to exclude other natural or anthropogenic thermal anomalies, the observed divergence in trends is less likely to be attributable to external heat sources. Instead, subsurface combustion within the coal seams is inferred to be the dominant driver of the significant LST trend divergence observed at coal-fire points, apart from natural seasonal fluctuations. This relationship is particularly evident during periods of intensified combustion, when the deviation between the STL trend of the coal fire and the nonfire trend increases proportionally, indicating the strengthening of fire activity over time.

To quantify the influence of coal-fire anomalies on the trend component, we derived an NFT, defined as ΔTrend. This was calculated by subtracting the trend of the corresponding synchronous background point from the STL trend component of each coal-fire sample point (Fig. 4). This metric effectively isolates coal-fire-related long-term change by removing background variability and periodic seasonal effects. The NFT exhibits substantial temporal variability over the monitoring period (2013–2024), with alternating phases of increase and decrease that are likely driven by a complex combination of climatic variability, mining disturbances, and fire-suppression interventions. Markedly, over the decade-plus monitoring window, more than 90% of coal-fire sample points have shown an overall declining NFT. The largest NFT reductions are observed in suppression sections A and C, confirming the success of these mitigation efforts. Temporally, the aggregated NFT predominantly increased prior to 2016 and shifted to a predominantly decreasing trajectory thereafter, although intermittent fluctuations persisted. Spatial heterogeneity is evident across the coalfield. Unmanaged or actively mined sections (notably in E, G, and F) exhibited rising or rebounding NFT, indicating that combustion activity persisted or intensified in these specific areas.

C. BFAST Abrupt Detection and Characteristics

Based on the modeling algorithm outlined in Section III-C, BFAST decomposition and breakpoint detection were applied to the net LST data cubes (LST_{net}) derived from background-corrected time-series pixels across the coalfield. The BFAST analysis extracted four key attributes: breakpoint timing, spatial location, magnitude of change, and segmented slope (Fig. 5), with results and statistical summaries presented in Figs. 6 and 7.

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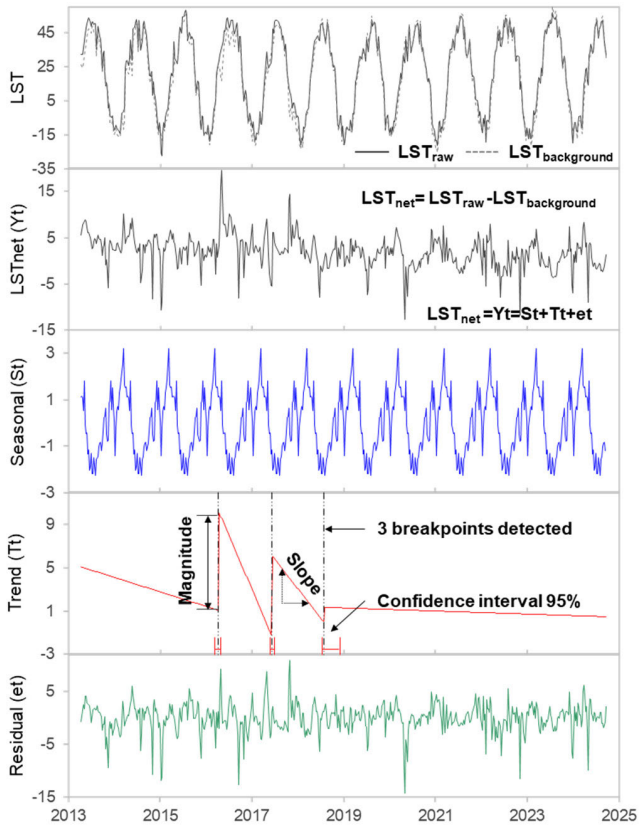


Fig. 5. BFAST time-series decomposition and attribute extraction for a representative LSTnet pixel. This procedure isolates nonstationary thermal shifts by decomposing background-corrected time series into seasonal, trend, and residual components. The extraction of breakpoints, magnitude, and slope from the trend provides the quantitative framework for characterizing abrupt transitions in fire dynamics and distinguishing combustion-driven signals from environmental noise.

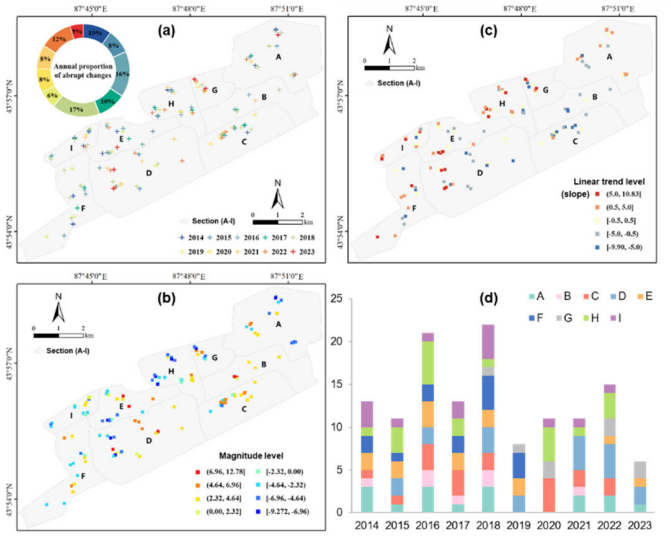


Fig. 6. Spatiotemporal distribution and interannual statistical attributes (d) of detected LST breakpoints. This figure maps the timing (a), magnitude (b), and segmented slope (c) trajectory of abrupt thermal shifts. It categorizes identified events into intensification and recovery phases, facilitating the spatial identification of high-frequency disturbance clusters across the monitoring period.

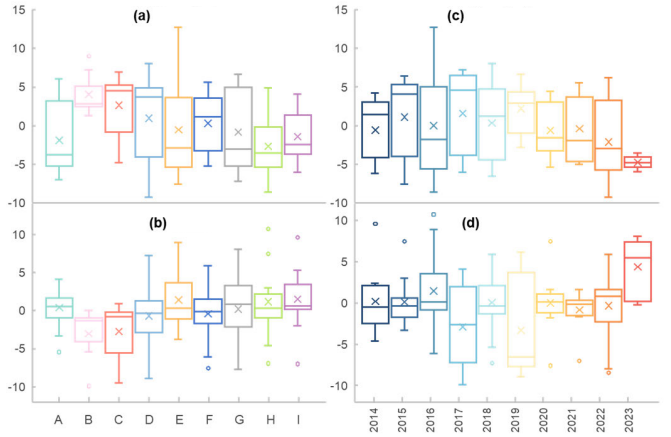


Fig. 7. Zonal statistical evaluation of breakpoint magnitudes and segmented slopes. This analysis aggregates change metrics across Miquan coalfield sections (A–I) to quantify thermal trajectories. It facilitates a comparative assessment of the variance between suppression-induced attenuation and mining-driven heat release. (a) and (c) Magnitude. (b) and (d) Slope.

From the spatial distribution of breakpoint timing of Fig. 6(a), high-density breakpoint clusters are predominantly concentrated in the western unsuppression sections (D–I), indicating persistent and detectable thermal disturbances in these areas. In contrast, breakpoints in the eastern suppression sections (A–C) are sparse and spatially dispersed, with significantly weaker clustering than in the west. Across the study area, approximately 150 trend breakpoints were detected between 2013 and 2024. Interannual statistics in Fig. 6(d) show that, except for fewer than three years, trend breakpoints occurred almost annually between 2014 and 2023, reflecting the widespread recurrence of high-frequency LST abruption events during this period. Specifically, peaks in abrupt events were observed in 2016 (16% of total) and 2018 (17%), whereas significant troughs occurred in 2019 (6%) and 2023 (5%). Notably, managed sections A and C exhibited marked gaps in abrupt changes between 2019 and 2020, likely reflecting the short-term suppression effects of local coal-fire governance and control policies.

Regarding the spatial intensity and breakpoint abruptions type of Fig. 6(b), the magnitude of LST abruptions across the coalfield ranges from −9.27 °C to 12.78 °C. Negative abruptions (magnitude < 0 °C, with post-change trends decreasing) account for 52%, while positive abruptions (magnitude > 0 °C, with post-change trends increasing) represent 48%. Spatially, positive abruptions are predominantly distributed in the eastern and southern regions (sections B, C, and D), whereas negative abruptions are more concentrated in the northwestern sections (D–I). Sections D and E exhibit the largest positive and negative extremes in abruption magnitude, along with the widest value ranges and strongest fluctuations in Fig. 7(a). These patterns are closely linked to intensive local open-cast mining activities. Regarding interannual variation in Fig. 7(c), the regional average magnitude of abrupt changes displays a trajectory of gradual increase followed by rapid decline. The cumulative increase from 2013 to 2019 was approximately

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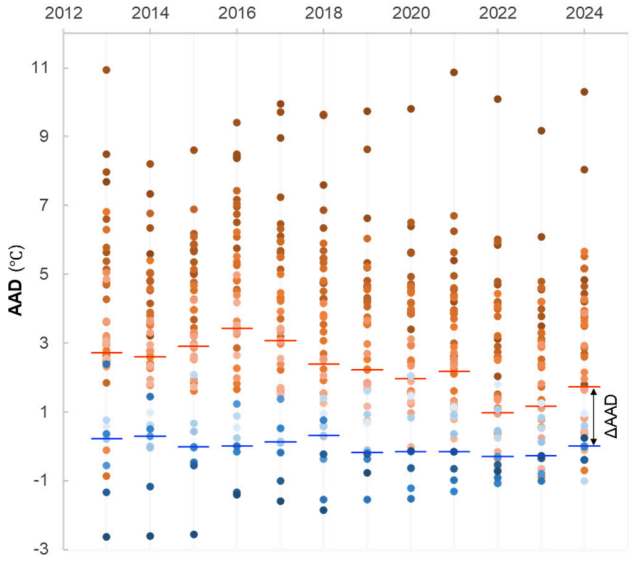


Fig. 8. Annual AAD distributions for series of sample points of coal-fire (brown) and noncoal-fire (blue). The red (blue) line indicates the first quartile (median) used as representative statistics for group-level comparison. This diagnostic plot establishes a quantitative benchmark for the thermal gap between combustion zones and the background. The statistical framework isolates persistent fire-driven thermal signals from shared climatic variability.

3 °C, whereas the cumulative decrease from 2020 to 2024 reached 5 °C.

Analysis of segmented trend slopes in Fig. 6(c) indicates a clear inverse relationship between slope values and the magnitude of abrupt changes. Spatially, significant negative slopes (Slope < 0; 33% of total), representing LST attenuation, are primarily distributed in the eastern coal-fire management zones (A–D). In contrast, strong positive slopes (Slope > 0; 34% of total), corresponding to LST intensification, are concentrated in the western mining and nonmanaged areas (E–I). Among all sections, the most pronounced negative and positive slopes occur in sections C and E, respectively, as shown in Fig. 7(b). This spatial contrast highlights two distinct processes: the progressive thermal recovery induced by fire suppression in managed areas, and the persistent or intensified heat release driven by mining activities in unmanaged areas. Temporally, the interannual evolution of regional mean slopes of Fig. 7(d) follows a two-stage trajectory: a gradual decline of about four units from 2013 to 2019, followed by a sharp rise exceeding five units after 2020.

D. Background Deviation and Classification of Coal-Fire Anomalies

The analysis of the AAD metric was crucial for examining the long-term temporal differentiation between coal-fire anomalies and background thermal conditions. The calculated AAD values for all coal-fire and nonfire background points were integrated into a temporal scatter plot (Fig. 8), revealing pronounced regional heterogeneity. The highest AAD values are concentrated in active mining sections such as A, D, and E, underscoring the strong linkage between coal mining activities and fire intensity (e.g., increased frequency and severity of spontaneous combustion). In contrast, sections C, B, and H exhibit the lowest AAD values, where the thermal regime is weakly affected by coal fires and dominated instead by periodic fluctuations related to suppression progress and natural climatic variability. Comparable spatial heterogeneity is also observed in background zones. In particular, extinguished coal-fire areas within sections B, D, E, and I display distinctly lower AAD values relative to other background areas, likely due to higher vegetation cover or proximity to water bodies.

Temporal analysis (Fig. 8) shows that the AAD values of coal-fire points (brown) exhibit an overall decreasing trajectory over the 17-year period. The sharpest reductions occurred in sections A and D within one to two years following the implementation of fire suppression. This highlights the pronounced time-dependent effectiveness of human intervention in mitigating fire activity. By contrast, the AAD values of noncoal-fire points (blue) remain relatively stable, yet display a gradual upward trend over time, suggesting a slow accumulation of thermal influence from adjacent coal fires on background areas. To statistically characterize the thermal contrast between the two populations, the first quartile (Q_1, red line) of annual AAD values for coal-fire points and the median (blue line) for noncoal-fire points were extracted as representative descriptive metrics. It is important to clarify that these metrics are not intended as universal thresholds for spatial separation, but rather as diagnostic indicators to provide an intuitive and quantitative benchmark for the "thermal gap" between combustion zones and their background. Quantitative assessment indicates that Q_1 of coal-fire anomalies declines over time (range: 0.97 °C–3.42 °C, mean: 2.28 °C), reflecting a progressive moderation in fire-related heat release. In contrast, the median of background AAD values remains near neutral (range: −0.31 °C–0.32 °C and mean: −0.02 °C), consistent with the thermal inertia of natural land surfaces. Subsequently, the annual ΔAAD, derived from $Q_1(\mathrm{AAD_{fire}})$–$\mathrm{Median}(\mathrm{AAD_{nonfire}})$, range from 1.28 °C to 3.42 °C, with a 17-year mean of 2.30 °C. This mean is consistent with the average LST trend deviation between coal fire and background areas (2.97 °C) reported for the Wuda coalfield [58], thereby supporting the validation of the ΔAAD metric, as a quantitative indicator, for evaluating the significant and persistent thermal contrast between coal-fire and background areas.

Finally, to characterize the statistical properties of the NFT, we applied the MK test (time step: one year) to assess its significance. The tau coefficient (trend direction), Z-value (trend strength), and p-value (significance level) were jointly utilized to evaluate the trend attributes. In parallel, the interannual change rate was estimated with Sen's slope (11), while the AAD of the NFT (12) was employed to quantify anomaly magnitude relative to the background. By integrating these four indicators—Z-value, p-value, Sen's slope, and AAD—a classification framework was developed (Table S5), categorizing fire pixels into four dynamic states.

Fig. 9 shows the temporal distribution of these states for all coal-fire pixels. The classification highlights the core features of fire evolution; while localized fire intensification is evident, the overall trend is dominated by decreasing

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Fig. 9. Classification and temporal distribution of dynamic states across all coal-fire sampling points. By integrating MK trend statistics (Z- and p-values), Sen's slope, and AAD metrics, this framework categorizes thermal trajectories into four distinct states. This multidimensional synthesis facilitates the identification of fire intensification, stability, or moderation.

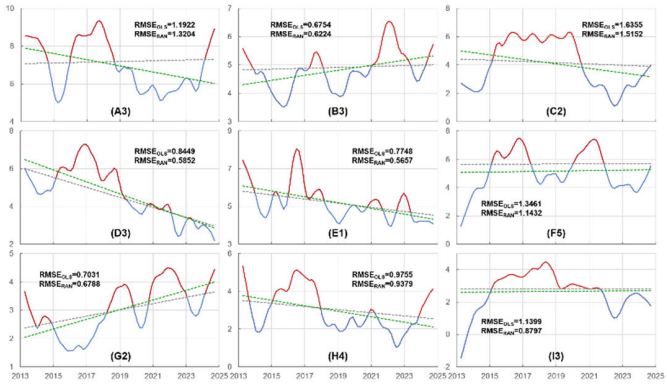


Fig. 10. RANSAC-based extraction versus OLS regression of net trend components for representative coal fire sample points in sections A–I. This figure illustrates the isolation of inlier background trajectories from outlier-driven fire anomalies. The RANSAC (gray dashed line) and OLS fitting bias (green dashed line) are compared. The method facilitates the binary classification of time-series segments into active burning (red) and nonfire background (blue) periods, providing the methodological basis for characterizing cyclical combustion dynamics.

states (35%). Increasing and stable states accounted for 33% and 17%, respectively, with the remaining 15% exhibiting nonsignificant patterns. This distribution closely matches the temporal dynamics of NFTs in Fig. 4. Notably, increasing fire states display strong spatial and causal associations with active mining zones (e.g., sections E–G), where sustained mining activities have repeatedly initiated or reactivated fires. These findings underscore the urgent need for targeted management strategies to curb fire expansion in such areas.

E. Extraction and SpatioTemporal Dynamics of Coal Fires

To characterize the temporal dynamics of coal-fire activity, the RANSAC algorithm (Section III-E) was employed for the robust fitting of long-term NFT components at the pixel scale. Fig. 10 illustrates representative fitting and detection results for selected coal-fire sample points across Sections A–I. In this framework, LST trend values exceeding the fit threshold (red segments) are identified as active fire periods, while values below the threshold (blue segments) correspond to nonsignificant fire states. By systematically labeling all pixels according to burning and nonburning periods, this approach enables the dynamic extraction of combustion cycles for coal-fire pixels

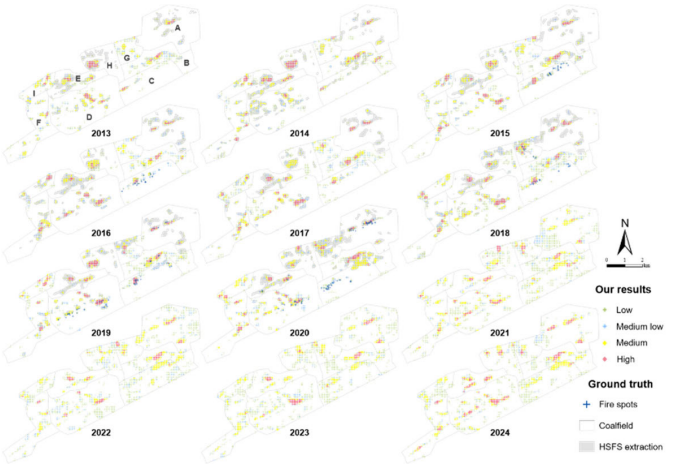


Fig. 11. Spatiotemporal distribution and verification of coal-fire frequency regimes. This framework categorizes thermal persistence into four probability levels based on RANSAC-extracted anomalies and a sparse sampling strategy. It facilitates the characterization of combustion stability and hazard gradients, providing the methodological basis for distinguishing sustained heat sources from transient or episodic signals.

across the entire study area. The results highlight that local coal-fire thermodynamic systems exhibit phased combustion behavior rather than continuous burning. Even confirmed fire points (e.g., A3–I3) show alternating burning and dormant intervals of variable length. These intermittencies essentially originate from the complex coupled dynamics of (sub)surface processes—driven by geological structure, oxygen supply, and human intervention. Such mechanisms impart pronounced spatiotemporal heterogeneity, resulting in nonstationary, cyclical fluctuations in heat release.

To evaluate the robustness of baseline extraction, we compared the RANSAC-based regression with conventional ordinary least squares (OLSs) fitting for representative coal-fire sample pixels in each region (Fig. 10). The results show that the total RMSE of OLS ($RMSE_{OLS}$), calculated over almost all observations, is consistently higher than the inlier-based RMSE of RANSAC ($RMSE_{RAN}$). Visually, the OLS fitting curve (gray) exhibits noticeable slope deviation in the presence of extreme thermal values, whereas the RANSAC curve (green) follows the normal fluctuation envelope and effectively suppresses anomaly-driven bias. These differences demonstrate that RANSAC provides a more stable estimation of background thermal evolution under coal-fire-induced disturbances.

The RANSAC algorithm was utilized to extract fire anomalies from pixel-level NFT components across the study area for 2013–2024. Spatiotemporal maps of coal-fire occurrence probability were generated using a sparse sampling strategy based on the maximum-value method within a 2×2 pixel window (Fig. 11). Crucially, rather than a binary fire mask, an evaluative framework was established to categorize the coalfield into four fire occurrence probability levels based on the persistence of burning-phase classifications: low (< 25%, green), medium-to-low (25%–50%, blue), medium (50%–75%, yellow), and high (> 75%, red). These levels represent a gradient of fire hazard and combustion stability. The defined percentages are

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not binary thresholds for fire detection, but rather represent the temporal frequency of burning-phase classifications over the observation period. Higher frequency values correspond to an elevated probability of sustained fire occurrence.

The extraction and classification in Fig. 11 show that high-frequency fire regimes (red pixels) display strong clustering and temporal persistence, spatially coinciding with the core zones of sustained thermal anomalies, thus representing long-term coal-fire heat sources amenable to long-term tracking. In contrast, low- (green) and medium-to-low-frequency (blue) regimes exhibit dispersed and transient patterns. These regimes (frequency $\leq$ 50%) are prone to periodic local adjustments in fire intensity and scale. Consequently, they exhibit rapid generation, propagation, and extinction patterns, reflecting the inherently unstable and episodic nature of coal-fire evolution. It is noteworthy that the extracted extent of fire-risk regimes exhibits an inverse relationship with frequency. Low-frequency fire zones cover the largest areas. However, as frequency increases, spatial dispersion diminishes and combustion becomes more spatially concentrated. This evolutionary pattern is highly consistent with the spatial attenuation of LST thermal field gradients with distance from the thermal source.

F. Accuracy Assessment and Validation

A verification benchmark was established based on historical records and field survey data. Spatial overlay comparisons were performed between field-measured fire-point locations, previously derived results (HSFS method), and the annual coal-fire-risk regimes in this study using the RANSAC algorithm (Fig. 11). The spatial correspondence of fire points was determined as follows. A measured fire point was defined as verified if it fell within the fire-hazard zones (encompassing all frequency levels) identified by the RANSAC algorithm. Table S6 reports the point-level validation results for measured fire locations (2015–2020). This site-level validation shows that RANSAC-derived fire-risk extents have strong spatial correspondence with observed fire points. Specifically, on average, 78% of measured fire points are located within areas exhibiting an annual occurrence frequency of $\geq$ 25% (classified as medium-to-low hazard and above). This indicates that confirmed combustion sources primarily correspond to regions with relatively stable thermal persistence throughout the year.

Range (area) validation was conducted by computing spatial overlap between the HSFS-derived polygons (gray areas in Fig. 11) and the RANSAC-derived fire-risk regimes. Validation results demonstrate high consistency between the two methods in both overlap rate (Table S7) and spatial distribution. The mean spatial overlap for fire-risk zones at medium-to-low frequency and above (> 25%) is 77% for 2013–2020, confirming the robustness of the RANSAC algorithm in capturing concealed combustion processes. Notably, in medium-to-high frequency zones (> 50%), the average spatial match exceeds 90%. Furthermore, as shown in Fig. 11, the HSFS method exhibits a systematic tendency to overestimate the fire extent, extracting larger spatial areas than those identified by RANSAC. This observation further validates the RANSAC algorithm's advantages in filtering transient noise, accurately delineating localized high-temperature coal-fire pixels. Consequently, the algorithm refines the spatial boundaries of active combustion hazard zones.

V. Discussion

A. Contribution and Innovation

This study proposes a robust framework for detecting and characterizing coal fires using long-term Landsat LST time series. By integrating extensive multiyear thermal observations, the framework moves beyond snapshot-based or short-term analyses and enables dynamic multitemporal characterization of coal-fire activity. Compared with traditional approaches that rely on limited datasets or fixed thermal thresholds, the use of long LST sequences substantially improves detection reliability and allows coal-fire evolution to be analyzed in a continuous and temporally coherent manner across heterogeneous fire zones.

As summarized in Table S8, the core methodological novelty of this study lies in the transition from traditional stationary thermal anomaly detection toward a dynamic, stage-wise evolution analytical framework. While long-term Landsat LST data have been widely used in previous studies, most approaches implicitly assume stationary or monotonic behavior and rely on empirically defined parameters. In contrast, coal-fire development is inherently nonstationary, involving ignition, expansion, suppression, and reactivation phases that manifest as structural changes in thermal signals. Our process-oriented design enables the identification of temporal transitions and fire evolution stages rather than merely detecting persistent anomalies, resulting in high-fidelity extraction of coal-fire regimes with validated spatial accuracy exceeding 77%.

A key contribution of this work is the explicit treatment of coal-fire thermal responses as nonstationary time series. Methodologically, the framework integrates STL decomposition, BFAST breakpoint detection, and RANSAC-based robust regression into a unified, stability-aware methodology, with explicit sensitivity and robustness evaluation of key parameters. STL is employed to separate seasonal fluctuations and disturbance noise from long-term trends, while BFAST captures peaks, troughs, and structural breakpoints in LST trajectories to identify both abrupt and gradual thermal changes. This nonstationary approach differentiates genuine combustion signals from solar-induced variability. Specifically, contextual normalization relative to local backgrounds cancels common-mode radiative heating, while STL separates the cyclical seasonal solar component from the persistent combustion trend. This ensures that the extracted hotspots represent structural transitions in thermal states rather than transient radiative fluctuations.

In parallel, the integration of STL decomposition with the MK trend test and Sen's slope provides a refined characterization of fire-induced thermal evolution. By separating long-term trends from seasonal fluctuations and disturbance noise, this approach allows the progression and regression of coal fires to be quantified dynamically. From a technical perspective,

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the framework enhances robustness in long-term anomaly extraction. The annual AAD, derived from decomposed trend components, quantitatively separates coal-fire-induced thermal anomalies from background variability, offering a potential reference for adaptive LST threshold definition. Moreover, as large fluctuations and outliers in long-term LST trends often distort conventional regression estimates, RANSAC-based robust regression mitigates the influence of anomalous observations caused by sensor artifacts or data gaps. This improves the stability and accuracy of background fitting in extended thermal time series.

Compared with conventional threshold- or gradient-based approaches, the proposed RANSAC–STL framework offers a distinct advantage by incorporating temporal structural consistency into boundary delineation. Traditional methods rely primarily on instantaneous thermal contrasts, which are sensitive to surface heterogeneity and secondary conductive heating. Such sensitivity often leads to boundary inflation in complex coalfield environments. In contrast, our approach evaluates whether each pixel exhibits a persistent structural deviation from its long-term climatic baseline, thereby distinguishing active combustion from background-coupled warming. This temporal constraint enhances spatial precision and reduces overestimation, particularly in regions characterized by strong thermal heterogeneity.

By explicitly evaluating the sensitivity and robustness of key parameters, the framework addresses the challenges of stochastic noise in long-term observations and ensures a more physically accurate and precise definition of the affected fire extents. Together, these elements constitute a comprehensive monitoring framework that goes beyond conventional LST-based anomaly detection methods by offering a transparent benchmark for interpreting long-term coal-fire lifecycle patterns.

B. Transferability and Applications

The proposed framework employs a pixel-level iterative time-series decomposition strategy to explicitly separate trend, seasonal, and residual components in long-term LST observations. The trend component captures persistent thermal evolution driven by climatic background, land management, and anthropogenic activities. In contrast, seasonal components represent periodic oscillations, while residuals mainly reflect transient weather effects and sensor-related noise. By isolating these components, the framework effectively mitigates temporal resolution limitations and random disturbances. It substantially reduces false fire detections and improves both the precision and spatiotemporal continuity of dynamic coal-fire monitoring. In this sense, the framework functions as a generalisable remote sensing analytical paradigm rather than a site-specific detection tool.

A key aspect of its transferability lies in the cross-disciplinary adaptation of time-series algorithms originally developed for vegetation dynamics. In ecological studies, STL and BFAST are widely regarded as standard tools for monitoring forest disturbance, recovery, and phenological shifts. By transposing these methods to subsurface thermal anomaly analysis, this study introduces a “Thermal Phenology” concept. Coal fires are interpreted as exhibiting phenology-like behaviors, including seasonal baseline oscillations, ignition-like intensification events, and suppression-driven attenuation phases. This conceptual transfer enables the identification of fire evolution stages instead of merely detecting static thermal anomalies, thereby providing a process-oriented understanding that stationary models cannot achieve.

Beyond coal-fire monitoring, the framework is inherently transferable to a broad range of environmental and ecological applications in mining and energy-affected regions. Its ability to separate long-term trends, seasonal variability, and transient disturbances makes it suitable for analyzing vegetation degradation, drought responses, ground deformation, and fire-related emissions. Owing to its scalability and reliance on widely available remote sensing time-series data, the approach can also be extended to other long-term landscape and environmental processes, such as geothermal anomalies, wildfires, volcanic activity, glaciers, and permafrost dynamics. It is also applicable to seasonal satellite products related to vegetation, land cover, and surface deformation.

C. Limitations

However, several data-, methodological-, and validation-related limitations should be acknowledged. At the data level, the absence of nighttime TIR observations in Landsat 8/9 imagery restricts the characterization of diurnal variations in coal-fire thermal radiation. The relatively long satellite revisit cycle, together with cloud contamination and quality-control filtering, further reduces the temporal completeness of the LST time series. Although interpolation was applied to maintain temporal continuity, missing or filtered observations cannot fully recover unobserved conditions, making the omission of short-lived thermal anomalies or rapid ignition and extinction events unavoidable. This limitation reflects the intrinsic constraint of satellite revisit frequency, which restricts the representation of subweekly thermal dynamics. In addition, the 30-m spatial resolution limits the detection of small or isolated fire spots and introduces uncertainty in precise fire-boundary delineation.

Meteorological filtering, while necessary to suppress cloud- and weather-induced artifacts, inevitably reduces data completeness and may remove certain short-term or marginal thermal variations. Although sensitivity analyses indicate that persistent combustion-driven signals are largely preserved, screening-induced discontinuities introduce uncertainty into reconstructed long-term thermal evolution, particularly when assessing subtle interannual variations. Despite the multistage filtering strategy, daytime solar radiation remains an unavoidable source of uncertainty in Landsat-based thermal anomaly detection. Topography-induced heating, terrain-shadow effects, and surface albedo variability may introduce localized thermal heterogeneity, especially where combustion signals are weak and comparable to background heating.

At the methodological level, uncertainties in atmospheric parameters associated with the RTE-based LST retrieval algorithm propagate into subsequent time-series analyses. The BFAST model rarely detects significant seasonal breakpoints

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in coal-fire regions and may exhibit limited sensitivity to gradual thermal transitions due to its piecewise linear segmentation assumption, potentially overlooking slowly evolving combustion processes. Furthermore, consecutive temporal gaps may amplify interpolation uncertainty during RANSAC-based fitting, influencing local trend estimation in sparsely observed periods. In addition, the framework (STL) assumes additive separability among trend, seasonal, and residual components. This assumption may introduce uncertainty under extreme surface disturbances or abrupt land cover changes, where component interpretation becomes less straightforward.

A further limitation arises from the translation between pixel-level anomaly detection and fire-area estimation. Fire-affected extents represent spatiotemporal aggregates of independently detected pixels rather than direct physical measurements. Because spatial coherence and temporal persistence jointly determine fire boundaries, uncertainty in individual pixel detection does not propagate linearly to fire-area uncertainty, making the relationship between parameter sensitivity and mapped fire extent inherently complex.

At the validation level, the availability of field-measured fire and background samples remains limited. Existing in situ observations are primarily concentrated during 2015–2020 and therefore do not fully encompass the complete fire evolution cycle extending to 2024. Moreover, deeply concealed coal fires producing subtle surface thermal responses have not been comprehensively validated against subsurface temperature measurements. These constraints may limit the spatial and temporal representativeness of performance evaluation, and results for recent years should therefore be interpreted with appropriate caution.

D. Future Research

From the data perspective, the integration of higher spatial-resolution and multitemporal TIR observations, particularly combined day–night measurements (e.g., UAV-based thermal imaging), would substantially improve the separation of combustion-driven signals from solar-induced radiative effects. The incorporation of multisource satellite observations, including Sentinel-3, GF-5, and MODIS products, could mitigate temporal gaps associated with individual sensors. This enables the construction of higher frequency and more continuous LST time series while maintaining acceptable noise levels. In addition, coupling thermal observations with auxiliary datasets such as geological information, meteorological records, and borehole temperature measurements would improve atmospheric correction and physical consistency in RTE-based temperature retrieval. The development of terrain–radiation correction approaches is also expected to reduce uncertainties caused by topography-induced heating and surface heterogeneity.

To address current validation limitations, future studies will prioritize the acquisition of updated field observations to verify post-2021 thermal evolution patterns. Planned efforts include the integration of geological surveys and drone-based thermal measurements for cross-validation of satellite-derived results. The deployment of fixed in situ thermal and gas-flux monitoring sensors is also envisaged to enable continuous ground-based observations, which strengthens the linkage between satellite-derived indicators and subsurface combustion dynamics.

From a methodological perspective, future developments should focus on advanced multisensor data fusion and spatiotemporal reconstruction. These algorithms should jointly exploit high-spatial and temporal-resolution observations. Beyond single-variable LST analysis, integrating complementary environmental indicators such as vegetation dynamics, surface deformation, land cover change, and gas emissions would enable investigation of the coupled evolution between coal fires and surrounding environmental systems. Such multidimensional monitoring frameworks will support improvement on early warning, risk assessment, and long-term fire mitigation strategies in complex mining environments.

VI. Conclusion

This study proposed a comprehensive framework for detecting and characterizing temporal changes in long-term thermal anomaly series, integrating time-series decomposition, breakpoint detection, and coal-fire extraction algorithms. The model was applied to the Landsat-derived LST time series from 2013 to 2024 to monitor and assess the persistent coal-fire dynamics in the Miquan coalfield, Xinjiang.

LST time series were first retrieved using the RTE algorithm in coal-fire zones, followed by the spatiotemporal analysis of LST distribution and differentiation patterns. The combined use of the STL decomposition and BFAST iterative model enabled the identification of temporal breakpoints in the trend component, thereby isolating abrupt shifts induced by coal-fire events. Through STL decomposition, seasonal, trend, and residual components were separated from the fluctuating LST time series. Subsequently, BFAST was employed to iteratively detect the timing, amount, and types of breakpoints. This allowed for the quantification of the magnitude and direction of thermal anomaly abruptness and statistical evaluation of divergence patterns across different spatiotemporal contexts.

Furthermore, the MK test and Sen's slope were utilized to evaluate the interannual trend and rate of LST change. The AAD index was also developed to quantitatively distinguish anomalous thermal intensity from the background baseline. The RANSAC algorithm was subsequently applied to extract potential fire pixels from the net trend component, enabling the spatiotemporal mapping of coal-fire evolution. Through the frequency analysis, four risk levels of coal-fire occurrence were classified across the study area. Validation with literature records and field investigations confirms that the proposed method achieved approximately 80% accuracy in fire-point identification and spatial matching, demonstrating its robustness and applicability.

The proposed framework demonstrates the advantages of long-term remote sensing data and advanced time-series algorithms in monitoring persistent and slowly evolving disasters. This overcomes the limitations of traditional finite sequence coal-fire detection. It ensures stable baseline fitting and significantly reduces false detection compared with conventional regression-based approaches. Despite certain limitations concerning data completeness, model applicability, and field

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validation, the method shows strong potential for further development. Future research should integrate multisource day–night thermal observations and advanced spatiotemporal data fusion to improve the resolution, information integrity, and physical consistency of long-term coal-fire monitoring. Strengthened field validation and multiindicator environmental analysis will further enhance early warning capability and support sustainable fire management in mining regions. Additionally, extending this approach to monitor multi-indicator phenological variables can facilitate long-term environmental evolution analysis and multidimensional risk assessment in mining areas.

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